

Sambandh

Core innovations, and the roadmap they make possible.

An evidence-grounded criminal network analysis platform correlating persons, finances and incidents across jurisdictions under auditable, role-based access.

THE ONE IDEA

The link is already in the data. It is not missing, it is unreachable.

Three FIRs in three police stations. One person in all three, accused in one, a witness in another, a phone contact in the third. That connection has been in the police database the whole time. Finding it currently needs an officer who happened to think of it and days of cross-referencing. Criminal networks are organised. The records are not.

HOW WE GOT HERE

The four gaps we identified

We did not start from what police data lacks. India digitised its police records years ago, and a solution premised on missing data would be unbuildable. We started from what a digitised record still cannot do, which is tell you what it connects to. That single reframe turned this from a search problem into a traversal problem, and a traversal problem has a structural answer rather than a better-interface answer.

GAP WE FOUND	WHAT IT ACTUALLY COSTS	WHAT WE BUILT AGAINST IT
Records are connected in reality but not in structure	A link that exists is found only if an officer thinks to look for it	A graph written automatically as cases are filed
The people who need the answer cannot write the query	Analytical capability stops at trained analysts in state headquarters	Conversational access by voice, in Kannada or English
An AI answer is unusable in an investigation unless every number is traceable	Fluent output that cannot be checked is worse than no output	The model selects verified tools and never computes
Nothing watches. Everything is reactive	A network is discovered after the fifth incident, not the first	Detectors that run unasked and raise standing alerts

THE DECISION WE REJECTED

The obvious build is to let the model write database queries. We rejected it for three reasons. It is unverifiable, because nobody can confirm a generated query expressed the question correctly and a subtly wrong join returns a confident wrong answer. It is unsafe, because a generated query is arbitrary code against a police database. And it is unbounded, because the model can express queries that cross access boundaries the system exists to enforce. So we inverted it: the model's entire authority is choosing from a menu we wrote and audited.

CORE INNOVATIONS

Seven capabilities, and where each one goes

Only the capabilities that are the solution, not the full feature list. Each is stated as it exists today, what is genuinely distinctive about it, the next development it makes possible, and why that matters operationally.

01 The self-building criminal network

TODAY A filed FIR, whether typed, scanned or handwritten in Kannada, is read, its people, places and accounts extracted, matched against persons already on record, and written into the relational, vector and graph stores in a single pass.

WHAT IS NEW ABOUT IT Nobody curates the network. Most graph systems fail because maintenance is manual and the model decays; this one has no maintainer, because the graph is a byproduct of filing a case.

WHERE IT GOES NEXT Call detail records as a second edge type. Police already do network analysis on CDR dumps by hand in spreadsheets, and it is the most-used investigative data in the country, entirely disconnected from case records today.

WHY IT MATTERS Network coverage grows automatically with case volume, at zero marginal analyst cost per district. That is what makes a rollout survivable.

02 A deterministic core under a conversational surface

TODAY An officer asks in plain language. The model selects from roughly forty audited tools, chains them across turns, and writes the answer around what those tools returned. It never writes a query and never computes a number.

WHAT IS NEW ABOUT IT Answers degrade toward the truth rather than away from it. Results and graphs render before any narration exists, so if the model fails halfway through, the analysis is already on screen. The analysis never depended on the AI.

WHERE IT GOES NEXT A per-claim evidence trail, so every sentence in an answer links to the exact tool call that produced it, turning a response into a citable document.

WHY IT MATTERS This is what makes an AI answer usable in a case file rather than merely persuasive.

03 Answers that are objects, not paragraphs

TODAY Every tool renders as a typed interactive component inline: a force-directed network, a hotspot map, a money-flow diagram, a risk card. Clicking a node opens entity actions and deep-links into the relevant workspace.

WHAT IS NEW ABOUT IT The investigator never types an identifier. Re-typing identifiers is exactly where real investigative software dies, and the widget contract is simultaneously the interface and the safety boundary that makes generated queries unnecessary.

WHERE IT GOES NEXT An investigation canvas: pin findings from across a conversation onto a case board and export it as a court-ready brief with the evidence trail intact.

WHY IT MATTERS The investigation flows continuously instead of restarting at every question.

04 Network intelligence

TODAY Community detection groups co-offenders into probable crews. PageRank identifies operational leadership, betweenness identifies the broker bridging two groups, and a trained model proposes plausible unconfirmed links, always labelled as leads.

WHAT IS NEW ABOUT IT In our own risk model, network centrality outweighs offence-severity history by roughly seven to one. Who a person offends with predicts more than what they have done, which justifies the graph approach from inside the model rather than from architecture.

WHERE IT GOES NEXT Watching the graph change shape rather than querying its current state, so the system can flag the moment one new edge merges two previously separate crews.

WHY IT MATTERS The difference between reconstructing a network after the fact and observing one form.

05 Financial trail intelligence

TODAY Four money-laundering typologies detected against transaction data and the account graph: structuring under the cash-reporting threshold, dormant funnel and mule accounts, layering through circular flows, and organised clusters of flagged accounts.

WHAT IS NEW ABOUT IT Deliberately rules and graph algorithms, not machine learning. An allegation of laundering has to trace to specific transaction identifiers, not to a score, so this is a case where knowing when not to use ML is the design decision.

WHERE IT GOES NEXT Joining the case graph to the financial graph at the person level, so who funds a crew becomes one question rather than two investigations.

WHY IT MATTERS Disrupts the economics of organised crime, not only its arrests. The same engine has direct commercial application in bank compliance.

06 Forecasting and proactive alerting

TODAY A self-exciting point process forecasts incident rate per square kilometre, and the same detectors that answer questions also run on a schedule and raise standing alerts without anyone asking.

WHAT IS NEW ABOUT IT The forecast separates chronic socio-economic baseline from acute near-repeat elevation, so the same number implies two different interventions. A chronically elevated area needs policy action; an acutely elevated one needs a patrol this week. A heatmap cannot tell you which.

WHERE IT GOES NEXT Continuous rather than scheduled detection, and extension of the same mathematics to public-order forecasting around large gatherings and festivals.

WHY IT MATTERS Prevention rather than reaction, and directed patrol coverage instead of broad-area patrolling, which is fewer vehicle kilometres for equivalent presence.

07 The trust layer

TODAY Seven roles as named capabilities, enforced inside the database rather than the interface, with field-level masking, an append-only record of every access, and model cards published inside the product.

WHAT IS NEW ABOUT IT A policy maker holds an orthogonal permission set, not a lower one: statewide aggregates yes, individual case data and the network graph never. That is the ecological fallacy prevented as code, so district-level correlation can never become an inference about a person.

WHERE IT GOES NEXT Hash-chaining each audit entry to its predecessor, making the record tamper-evident and extending it into a digital evidence chain of custody.

WHY IT MATTERS This is the layer that determines whether a system this capable is deployable in a democracy at all.

VISION

Three rungs, each resting on something already working

A roadmap is only credible if every step is a consequence of what already exists. Each rung below names what we would build, what we would train, and the part we consider genuinely hard.

01 Call detail records: the data police already use

TODAY Communication becomes a third edge type alongside co-offending and financial links.

WHAT IS NEW ABOUT IT It is deliberately humble, and that is the point. CDR analysis is what Indian police do today, manually, in Excel. Building for the real workflow rather than an imagined one.

WHERE IT GOES NEXT Initially nothing is trained; the value is the data. Then link prediction is retrained on the richer multiplex graph, where communication edges are a strong association signal.

WHY IT MATTERS Honest hard part: CDR volume dwarfs case data, so this becomes a scale and retention policy question rather than a modelling one.

02 The network as a live instrument

TODAY Recompute only the neighbourhood an incoming FIR touches, and compare the network's structure before and after, watching for structural events rather than thresholds.

WHAT IS NEW ABOUT IT The signal we want is the moment one new edge merges two clusters that were separate. That is a bridge forming between two crews, one of the highest-value events in organised crime, and nobody detects it today because existing tools query a graph rather than watch it change shape.

WHERE IT GOES NEXT Nothing is trained. This is an algorithms problem, and incremental community detection is established work rather than speculation. It rests on two things already built: community detection, and alerts fingerprinted by group membership rather than by the algorithm's arbitrary label, which is exactly what makes a merge detectable as an event rather than dismissible as relabelling noise.

WHY IT MATTERS Honest hard part: calibrating which new edges are meaningful, which needs real confirmed cases.

03 Working across states without pooling data

TODAY States will not surrender raw crime records to a central pool, and that is law and politics rather than a technical gap. So the national system cannot be one database.

WHAT IS NEW ABOUT IT Two separate mechanisms. Federated query answers whether a person appears in another state's records, with only a match returning and never a record. Federated learning trains shared models where only weights are exchanged. We already isolate by district, so this is our own principle one level up.

WHERE IT GOES NEXT We would federate modus operandi linkage first. It is gradient-based so weight averaging applies directly, a burglary signature is genuinely the same behaviour in every state, and it reads only method and never identity, so the privacy case is clean. Link prediction federates partially, since the graph embedding is local but the classifier above it is not. The survival model is the awkward one, because tree ensembles do not average like gradients do.

WHY IT MATTERS Honest hard part: matching an identity across states without exchanging the identity. The relevant techniques are hashed identifier matching and private set intersection.

AND THEN IT COMPOUNDS

Today our models are trained offline and frozen. Every lead an investigator confirms or dismisses is a labelled example that is currently discarded. Capturing that judgement turns usage into training data, and the system improves because it is used rather than despite it. It also produces exactly the per-district record a real fairness audit requires.

COMPARISON

Three different classes of alternative

ALTERNATIVE	WHAT IT IS	WHERE IT STOPS	WHAT WE DO INSTEAD
IBM i2, Palantir	Analyst workbenches for link analysis	A trained specialist assembles the chart manually, at significant licence cost, in English, on a desktop	The network assembles itself, and an investigating officer queries it by speaking Kannada
Text-to-SQL and text-to-Cypher tools	A language model generates a database query from a question	The query is unverifiable and unsafe, and an unverifiable answer is worse than none in an investigation	We permit no generated queries at all
CCTNS, ICJS	The national systems of record for police data	They store cases well and connect them badly; they were not designed to correlate	Not a competitor but our substrate. We consume what they already produce

CONVICTION

Challenges we hit, and what we did about them

Five real problems from the build, each with the fix. These are the answers to give when asked what was difficult, because each one shows the difference between running an algorithm and shipping a product built on one.

THE CHALLENGE	HOW WE ADDRESSED IT
The assistant produced fluent, confident numbers that were not real.	We removed its ability to compute at all. It selects a verified tool and narrates the result. This became the architecture rather than a guardrail on top of it.
Alerts turned into noise. The same criminal crew re-alerted every hour.	The clustering algorithm rennumbers communities on every run, so keying on its label meant every rescan looked new. We fingerprint the sorted membership of the group instead, giving it a stable identity independent of the algorithm.
District access rules silently switched off inside streamed responses.	The database session context is transaction-scoped, and a streaming response returns before its body runs, committing and clearing it. Found because a case workspace came back empty only inside the assistant. Fixed by re-establishing the context inside the stream.
Financial detectors were too loose to be credible.	Funnel-account detection flagged any busy account. We added the prior-dormancy requirement the typology actually specifies, which is what makes the word dormant mean something, and validated all four against labelled ground truth.
We have no access to real crime records.	We generated a synthetic dataset with known ground truth so correctness could be measured rather than asserted: real districts, NCRB-derived socio-economic priors, persistent co-offending crews so the network has genuine structure. We report the weak results alongside the strong ones.

WHAT THE NUMBERS PROVE, AND WHAT THEY DO NOT

Our models are trained on synthetic data with known ground truth. Risk scoring reaches a held-out concordance of 0.71 against a 0.50 baseline. Link prediction reaches 0.77 AUC on a temporal holdout, trained on the graph as it stood at a cutoff and scored on edges that actually formed afterwards, rather than on a random split that would leak future structure. Modus operandi linkage reaches 0.90 recall at three. What this proves is that the methods work and the pipeline is correct. What it does not prove is generalisation to real records. That requires a district data-sharing agreement, and it is the first thing we would ask for.